Deliverable #1: Project Description & Entity-Relationship Diagram (ERD)

Project 25: Independent Living Support for Elderly People

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Project Description

Sanad Home Care Services operates an independent living support platform in Cairo for elderly people who live alone while their children live in other cities or abroad. The platform combines non-camera home sensors, an Arabic smart speaker, field caregivers, and a family dashboard to keep seniors safe without compromising their privacy. Everything is stored in one unified database covering the following business rules:

- **Members and Routine Profiles:**

Each elderly **Member** registered in the system has a unique member_id, full_name, date_of_birth, a derived age, phone_number, health_contact (doctor or clinic contact), one or more preferred_languages (multivalued, used for caregiver matching), and a

composite address (street, district, city). Each member's daily routine profile is stored through their usual wake_up_time (used as the morning inactivity cutoff), one or more meal_times (multivalued), and a composite multivalued medication_schedule (medicine_name, reminder_time, instructions) played aloud by the smart speaker. To protect privacy, each member record tracks a composite privacy_settings state consisting of monitoring_status (Active or Paused for up to 24 hours), pause_expiry_time, and away_mode_status (enabled when traveling so empty-home alarms are suppressed).

- **Informed Consent History:**

Before monitoring begins, a member must give informed consent in a video call with the Care Coordinator, and consent is reviewed every 6 months. Each **Consent** record is a weak entity identified by consent_date (date and time) within its owner Member, and stores consent_status (Pending, Active, Expired, Withdrawn), review_date, expiry_date, and video_call_uri. Every member has at least one consent record over time. Review_date is when the next 6-month review is due; expiry_date is when consent lapses if not renewed.

- **Family Contacts and Voice Messages:**

Each **Family_Contact** has a unique contact_id, full_name, phone_number, email, and local time_zone (e.g., Dubai or Canada, used to adjust daily summaries to local time). Each member has one or more family contacts, and a family contact may be linked to one or more members. For each link between a member and a family contact (**Has_Contact**),

the system stores the relation (e.g., Daughter, Son) and the call_order (priority order for anomaly escalations). Family contacts can also send voice messages (**Sends_Message**) to a member's smart speaker; each message interaction records sent_timestamp, audio_uri, and delivery_status. Each voice message is

stored as its own record, so a family contact can send many messages to the same member.

- **Home Hubs and IoT Sensors:**

Each member home has at most one installed **Home_Hub** connecting 6 to 10 devices. Each hub has a unique hub_id, installation_date, battery_status (0–100% supporting an 8-hour backup during power cuts), power_status (MainsAvailable or PowerOutage), connection_status (Online or Offline), and composite speaker_settings (voice_speed and voice_pitch for the Arabic smart speaker). Each installed **Sensor** has a unique sensor_id, a unique hardware serial_number, sensor_type (Motion, Door, or FallDetector), room location (e.g., Kitchen, Bathroom, Front Door), installation_date, sensor_status (Active, Inactive, Faulty, Offline), and wear_status (Worn, Unworn, Charging, or N/A for non-wearable sensors—used to trigger daytime speaker reminders if the fall detector is left charging on a table). Each hub connects to several sensors, and every sensor belongs to exactly one hub.

- **Sensor Events, Alerts, and Escalations:**

Each sensor reports **Sensor_Event** records over time, identified by event_id, with event_type (MotionDetected, DoorOpened, FallDetected, FallDetectorUnworn), event_timestamp, and event_details. When an unusual event occurs (such as a door opening at 3:00 AM, a hard fall, or no kitchen movement by the morning cutoff time), the system creates an **Alert** identified by alert_id. Each alert belongs to one member and may be triggered by a sensor event (or generated directly by a system rule, such as morning inactivity or a low hub battery). An alert records alert_type, alert_timestamp, speaker_response (ConfirmedFine within 5 minutes, NoAnswer, or NotApplicable), alert_status (Open, Escalated, Resolved, Closed), escalation_level (0 = Speaker, 1 = Family

Called, 2 = Caregiver/Volunteer Dispatched, 3 = Emergency 123), outcome (FalseAlarm, FamilyResolved, CaregiverHandled, EmergencyDispatched), resolution_time, and a composite emergency_dispatch record (dispatch_time, service_number, dispatch_status) for hard falls requiring emergency contact within 2 minutes. Each escalation step records which contact or responder was reached, when, and whether they answered.

- **Caregivers and Neighbor Volunteers:**

Each **Caregiver** has a unique caregiver_id, full_name, phone_number, district, daily_working_hours (Hours worked per day; capped at 10), availability_status, bg_check_expiry_date (renewed yearly), multiple care skills (multivalued: shopping, cleaning, companionship, mobility), and multiple spoken languages (multivalued). For urgent escalated alerts where family contacts do not answer within 10 minutes, the nearest available caregiver is dispatched (**Dispatched_To**, recording arrival_time to track the 45-minute Cairo target). For non-urgent alerts, verified **Neighbor_Volunteer** participants (identified by volunteer_id, with full_name, phone_number, district, and verification_status) may respond (**Assists_With**).

- **Scheduled Care Visits:**

Family contacts (or the Caregiver Lead) book scheduled **Visit** records for members. Each visit has a unique visit_id, visit_date, composite scheduled_time (start_time, end_time), service_type (Shopping, Cleaning, Companionship), visit_status, advance_notice_sent (confirming the 24-hour speaker reminder), composite verification_codes (check_in_code, check_out_code), composite actual_timestamps (check_in_time, check_out_time), a concise visit_note submitted by the caregiver, and a family_rating (1–5 score used to evaluate caregiver feedback). Each visit is assigned to one caregiver,

who performs it and submits the visit_note.

- **Wellbeing Reports (Daily & Monthly):**

The system generates **Wellbeing_Report** records for each member, modeled as a weak entity identified by report_date and report_type (Daily dashboard summary or Monthly 3-month wellbeing trend report) within its owner Member. Each report stores safety_status (Green/Normal or Red/Attention), sleep_status, meal_status, visit_completion_status, and activity_trend.

- **Subscriptions, Invoices, and Caregiver Payroll:**

A family contact subscribes for a member's care plan. Each **Subscription** has a unique subscription_id, plan_type, start_date, monthly_fee, and status. Each subscription generates monthly **Invoice** records identified by invoice_id, with issue_date, total_amount, and payment_status. Sanad also tracks caregiver **Payroll** records identified by payroll_id, storing pay_period, total_hours_worked, amount_paid, and payment_date. Each payroll record belongs to one caregiver and totals the hours from that caregiver's completed visits in the pay period.

Entities

#	Entity Name	Why It Is an Entity
1	Member	The elderly person living independently and receiving Sanad care services.
2	Consent	Historical record of a member's video consent and mandatory 6-month renewals.
3	Family_Contact	A relative or trusted person who receives alerts, books visits, and pays subscriptions.
4	Home_Hub	The central battery-backed in-home unit with an integrated Arabic smart speaker.
5	Sensor	An individual non-camera IoT sensor (motion, door, or wearable fall detector) in a home.
6	Sensor_Event	A raw activity or status reading reported by an installed sensor at a specific timestamp.
7	Alert	An anomaly or emergency event requiring speaker check-in, tiered escalation, or 123 dispatch.
8	Caregiver	A trained Sanad field employee who performs scheduled visits and urgent alert dispatches.
9	Audio_Message	Stores voice messages sent by family contacts to a member's smart speaker
10	Neighbor_Volunteer	A verified neighbor who responds to non-urgent community alerts (5-member team scope).
11	Visit	A scheduled in-home care session performed by a caregiver for a member.
12	Wellbeing_Report	A daily red/green dashboard summary or monthly wellbeing activity trend report for a member.
13	Subscription	A service plan covering a member and paid for by a family contact.
14	Invoice	A billing statement issued for a subscription.
15	Payroll	A salary payment record calculated from a caregiver's completed working hours.

Attributes

Entity	Attribute	Type	Note
Member	`member_id`	**Key**	Unique identifier for the elderly member.
	`full_name`	Simple	Member's full name.
	`date_of_birth`	Simple	Birth date used to calculate age.
	`age`	**Derived**	Calculated from '`date_of_birth`'
	`phone_number`	Simple	Member's contact number.
	`health_contact`	Simple	Member's doctor or clinic contact
	`address`	**Composite**	Components: (`street`, `district`, `city`)
	`preferred_languages`	**Multivalued**	Languages spoken by the member for caregiver matching
	`wake_up_time`	Simple	Usual morning wake-up cutoff time for the Morning Inactivity rule
	`meal_times`	**Multivalued**	Usual daily meal times
	`medication_schedule`	**Composite Multivalued**	Components: (`medicine_name`, `reminder_time`, `instructions`)
	`privacy_settings`	**Composite**	Components: (`monitoring_status`, `pause_expiry_time`, `away_mode_status`)
Consent *(Weak)*	`consent_date`	**Partial Key**	Timestamp when consent was recorded with the Care Coordinator.
	`consent_status`	Simple	Values: Pending, Active, Expired, Withdrawn
	`review_date`	Simple	Mandatory 6-month consent review date

	`expiry_date`	Simple	Date when current consent expires if unrenewed.
	`video_call_uri`	Simple	Reference to the renewal video call recording
Family_Contact	`contact_id`	**Key**	Unique identifier for the family member.
	`full_name`	Simple	Family member's full name.
	`phone_number`	Simple	Phone number used for escalation calls.
	`email`	Simple	Email address for account notifications.
	`time_zone`	Simple	Relative's local time zone
Home_Hub	`hub_id`	**Key**	Unique identifier for the home hub device.
	`installation_date`	Simple	Date the hub was installed by the Care Coordinator.
	`battery_status`	Simple	Backup battery charge percentage
	`power_status`	Simple	Values: MainsAvailable, PowerOutage
	`connection_status`	Simple	Values: Online, Offline.
	`speaker_settings`	**Composite**	Components: (`voice_speed`, `voice_pitch`)
Sensor	`sensor_id`	**Key**	Unique identifier for the sensor.
	`serial_number`	Simple	Manufacturer hardware serial number
	`sensor_type`	Simple	Values: Motion, Door, FallDetector
	`room`	Simple	Room location
	`installation_date`	Simple	Date the sensor was installed.
	`sensor_status`	Simple	Values: Active, Inactive, Faulty, Offline

	wear_status	Simple	Values: Worn, Unworn, Charging, N/A
Sensor_Event	event_id	**Key**	Unique identifier for the raw sensor reading.
	event_type	Simple	Values: MotionDetected, DoorOpened, FallDetected, PendantUnworn.
	event_timestamp	Simple	Date and time the sensor event occurred.
	event_details	Simple	Additional sensor reading metadata
Alert	alert_id	**Key**	Unique identifier for the generated alert
	alert_type	Simple	Values: HardFall, MorningInactivity, UnusualNightActivity, HubLowBattery, UnwornPendant
	alert_timestamp	Simple	Date and time the alert was triggered.
	speaker_response	Simple	Member's voice check-in response within 5 mins: ConfirmedFine, NoAnswer, NotApplicable
	alert_status	Simple	Values: Open, Escalated, Resolved, Closed.
	escalation_level	Simple	Values: 0 (Speaker), 1 (Family Contact), 2 (Caregiver/Volunteer), 3 (Emergency 123)
	outcome	Simple	Values: FalseAlarm, FamilyResolved, CaregiverHandled, EmergencyDispatched
	resolution_time	Simple	Timestamp when the alert was resolved (used with ‘alert_timestamp’ to compute average response speed)
	emergency_dispatch	**Composite**	Components: (dispatch_time, service_number , dispatch_status)
Caregiver	caregiver_id	**Key**	Unique identifier for the caregiver.
	full_name	Simple	Caregiver's full name.

	phone_number	Simple	Contact number.
	district	Simple	Assigned district for proximity matching
	skills	**Multivalued**	Care skills such as Shopping, Cleaning, Companionship, Mobility
	languages	**Multivalued**	Spoken languages such as Arabic, English, French
	daily_working_hours	Simple	Tracked daily hours
	availability_status	Simple	Values: Available, OnVisit, Unavailable, OnLeave.
	bg_check_expiry_date	Simple	Annual background check renewal date
Audio_Message	message_id	**Key**	Solid underline. Unique identifier for the voice message
	audio_uri	Simple	Secure storage reference/URI for the audio recording.
	sent_timestamp	Simple	Date and time the family contact sent the message.
	delivery_status	Simple	Values: `Queued`, `Sent`, `Delivered`, `Played`, `Failed`
Neighbor_Volunteer	volunteer_id	**Key**	Unique identifier for the neighbor volunteer
	full_name	Simple	Volunteer's full name.
	phone_number	Simple	Volunteer's contact number.
	district	Simple	District for proximity matching.
	verification_status	Simple	Values: Pending, Verified, Inactive.
Visit	visit_id	**Key**	Unique identifier for the scheduled caregiver visit.
	visit_date	Simple	Scheduled calendar date of the visit.
	scheduled_time	**Composite**	Components: (start_time,

			`end_time`)
	`service_type`	Simple	Values: Shopping, Cleaning, Companionship
	`visit_status`	Simple	Values: Scheduled, Confirmed, InProgress, Completed, Cancelled, NoShow.
	`advance_notice_sent`	Simple	Boolean tracking the mandatory 24-hour smart speaker advance notice
	`verification_codes`	**Composite**	Components: (`check_in_code`, `check_out_code`)
	`actual_timestamps`	**Composite**	Components: (`check_in_time`, `check_out_time`)
	`visit_note`	Simple	Concise structured post-visit note
	`family_rating`	Simple	1–5 rating from family
Wellbeing_Report *(Weak)*	`report_date`	**Partial Key**	Date or month covered by the report.
	`report_type`	**Partial Key**	Values: `Daily` (dashboard summary) or `Monthly` (3-month trend report).
	`safety_status`	Simple	Overall indicator: Green (Normal), Red (Attention/Emergency)
	`sleep_status`	Simple	Summary of sleep routine (Normal, Irregular).
	`meal_status`	Simple	Summary of kitchen/meal activity (AsExpected, MissedActivity).
	`visit_completion_status`	Simple	Summary of daily/monthly caregiver visits.
	`activity_trend`	Simple	Comparison of activity changes over the last 1–3 months
Subscription	`subscription_id`	**Key**	Unique identifier for the member's subscription plan
	`plan_type`	Simple	Tier of care package (e.g., Basic Monitoring, Full Care).

	`start_date`	Simple	Subscription activation date.
	`monthly_fee`	Simple	Monthly recurring charge.
	`subscription_status`	Simple	Values: Active, Paused, Cancelled.
Invoice	`invoice_id`	**Key**	Unique identifier for the billing invoice
	`issue_date`	Simple	Date the invoice was generated.
	`total_amount`	Simple	Total billed amount
	`payment_status`	Simple	Values: Paid, Unpaid, Overdue.
Payroll	`payroll_id`	**Key**	Unique identifier for the caregiver payroll record
	`pay_period`	Simple	Month and year of the payroll cycle.
	`total_hours_worked`	Simple	Total verified hours from completed visits.
	`amount_paid`	Simple	Total salary payout for the period.
	`payment_date`	Simple	Date the payout was issued.

Relationships

#	Name	Entities	Cardinality	Business Rule & Source Justification
1	**Gives_Consent** *(Identifying)*	`Member` — `Consent`	`Member` **1** : **N** `Consent`	Every member must give consent before monitoring begins and renew it every 6 months; a consent record cannot exist without its member (`REQ-01`).
2	**Has_Contact**	`Member` — `Family_Contact`	`Member` **M** : **N** `Family_Contact`	Every member has one or more family contacts called in priority

				order (`call_order`); a family contact can monitor one or more elderly relatives (`00 Project Request`).
3	**Sends_Audio**	`Family_Contact` — `Audio_Message`	`Family_Contact` **1**:**N** `Audio_Message`	Families can send voice messages that the speaker plays aloud.
4	**Has_Hub**	`Member` — `Home_Hub`	`Member` **1**:**1** `Home_Hub`	A member home has at most one installed hub (single line during onboarding before installation); every installed hub belongs to exactly one member.
5	**Connects_To**	`Home_Hub` — `Sensor`	`Home_Hub` **1**:**N** `Sensor`	Each home hub connects 6 to 10 sensors; every sensor must be connected to exactly one home hub (`05 Device Notes`).
6	**Reports**	`Sensor` — `Sensor_Event`	`Sensor` **1**:**N** `Sensor_Event`	A sensor may report many events over time (or none yet if newly installed); every sensor event is generated by one specific sensor.
7	**Triggers**	`Sensor_Event` — `Alert`	`Sensor_Event` **1**:**N** `Alert`	Most sensor events are normal routine movements and trigger no alert; also, some alerts (like Morning Inactivity) are triggered by time cutoff rules without a sensor

				event.
8	**Concerns**	`Member` — `Alert`	`Member` **1** : **N** `Alert`	A member may experience several alerts or none; every alert is associated with exactly one member.
9	**Dispatched_To**	`Caregiver` — `Alert`	`Caregiver` **1** : **N** `Alert`	Only urgent alerts that remain unanswered by the member (5 min) and family (10 min) dispatch a caregiver (`REQ-04`).
10	**Assists_With**	`Neighbor_Volunteer` — `Alert`	`Neighbor_Volunteer` **1** : **N** `Alert`	Verified neighbor volunteers respond only to non-urgent community alerts (`00 Project Request`).
11	**Books_Visit**	`Family_Contact` — `Visit`	`Family_Contact` **1** : **N** `Visit`	Families book visits for their parent (`00 Project Request`); partial on `Visit` because the Caregiver Team Lead can also schedule visits (`Workflow 2`).
12	**Receives_Visit**	`Member` — `Visit`	`Member` **1** : **N** `Visit`	A member may receive many scheduled visits or none yet; every visit is scheduled for exactly one member.
13	**Performs_Visit**	`Caregiver` — `Visit`	`Caregiver` **1** : **N** `Visit`	Every scheduled visit is matched and assigned to one caregiver; a caregiver performs many visits over time (`REQ-09`).

14	**Has_Report** *(Identifying)*	Member — Wellbeing_Report	Member **1** : **N** Wellbeing_Report	Daily dashboard summaries and monthly wellbeing reports exist solely for a specific member (REQ-11, 00 Project Request).
15	**Subscribes_For**	Family_Contact — Subscription	Family_Contact **1** : **N** Subscription	"A family member subscribes for a parent" (00 Project Request); one family contact pays for the subscription while other relatives may only be emergency contacts.
16	**Covered_By**	Member — Subscription	Member **1** : **1** Subscription	Every registered member is covered by one active subscription plan, and each subscription covers one member.
17	**Billed_Via**	Subscription — Invoice	Subscription **1** : **N** Invoice	A subscription generates multiple invoices over time; every invoice bills one specific subscription (00 Project Request, Workflow 2).
18	**Paid_Via**	Caregiver — Payroll	Caregiver **1** : **N** Payroll	Caregivers receive payroll records across pay periods; every payroll record belongs to one caregiver (00 Project Request, Workflow 2).
19	**Receives_Audio**	Member – Audio_Message	Member **1**:**N**	A member may

			`Audio_Message`	receive many voice messages or none; every voice message is directed to exactly one member